# Does More Complex Mean More Accurate?

## Interpretable NBA Player Representation and Next-Season Forecasting with PlayerMatch

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**Abstract**

Player forecasting in basketball creates an intuitive temptation: if player development is nonlinear and context dependent, a more flexible machine-learning model should outperform a simpler one. This paper tests that assumption using PlayerMatch, an interpretable NBA analytics system that combines standardized player representations, weighted cosine similarity, principal component analysis, k-means clustering, historical profile tracking, and next-season forecasting. The forecasting experiment uses 1,651 eligible player-season transitions from 2020–21 through 2025–26 and compares a persistence baseline, Ridge regression, and Gradient Boosting for points, rebounds, assists, true shooting percentage, and usage rate. Model families are selected using expanding-window validation with a pre-specified 2% simplicity tolerance, while 2025–26 is held out from model selection and used only for final evaluation. Ridge regression is selected for all five outcomes. Relative to persistence, the held-out MAE improves by 7.35% for points, 5.95% for rebounds, 1.67% for assists, 10.87% for true shooting percentage, and 7.50% for usage. The aggregate result is not uniform, however. For top-decile scorers, the true-shooting model performs worse than persistence and underpredicts by 1.04 percentage points on average. A Shai Gilgeous-Alexander case study illustrates this subgroup failure, while a Gradey Dick–De'Andre Hunter case study shows how statistical similarity changes over time and across weighting choices. The results suggest that, for this data and forecasting problem, additional model complexity does not automatically produce meaningful out-of-sample improvement, and aggregate accuracy can conceal player-type-specific weaknesses.

## 1 Introduction

Basketball analytics often moves between two different questions. The first is descriptive: *what type of player is this, and who plays a statistically similar role?* The second is predictive: *what should we expect this player to produce next season?* Both questions matter for scouting, roster construction, player development, and public-facing analysis, but they require different statistical tools and different standards of evidence. A similarity model can be useful even if it is not a forecast. A forecast, by contrast, should be judged against a credible baseline on data that were not used to choose the model.

PlayerMatch was built to connect these two sides of analysis in one interpretable workflow. It represents players using ten season-level statistics, standardizes those statistics relative to the league, allows alternative feature-weighting profiles, measures pairwise similarity with weighted cosine similarity, and provides two-dimensional principal-component maps and k-means archetypes for exploratory analysis. A separate supervised pipeline uses consecutive player seasons to forecast

five next-season outcomes. The application is therefore not intended to reduce a player to a single rating. Its purpose is to make several statistical views of the same player available while keeping the assumptions behind each view explicit.

The forecasting component created the main empirical question for this paper. Before running the experiment, I expected Gradient Boosting to perform somewhat better than Ridge regression. The expectation was reasonable: player development can involve nonlinear relationships among age, role, playing time, usage, efficiency, and recent changes in production. I did not expect the difference to be enormous, but I expected the more flexible model to have at least some advantage. The experiment did not support that expectation strongly enough to justify the extra complexity.

The primary research question is:

> *To what extent do more complex statistical models improve next-season NBA player forecasts over simpler persistence and regularized linear approaches?*

A secondary question is:

> *Does the amount of predictive improvement vary depending on the statistic being forecast and the type of player being evaluated?*

These questions are deliberately narrower than asking for the "best" machine-learning model in basketball. The answer is conditional on the available data, the prediction horizon, the eligible-player definition, the engineered features, the evaluation window, and the error metric. This paper therefore treats complexity as something that must earn its place through out-of-sample performance rather than as a goal in itself.

The main result has three parts. First, Gradient Boosting occasionally records the lowest expanding-window validation error, but its advantage is small. For points and true shooting percentage, Ridge lies within 0.55% and 0.02%, respectively, of the raw validation-best Gradient Boosting model. Under a pre-specified rule that prefers a simpler candidate when it is within 2% of the best validation MAE, Ridge is selected. Ridge is also the raw validation-best model for rebounds, assists, and usage rate. Second, the value of forecasting differs substantially across targets: the held-out improvement over persistence ranges from only 1.67% for assists to 10.87% for true shooting percentage. Third, overall error hides meaningful subgroup behavior. The true-shooting model improves strongly in aggregate but loses to persistence among top-decile scorers and exhibits a negative mean signed error for that group.

The descriptive side of PlayerMatch provides a complementary result. Gradey Dick and De'Andre Hunter show a steady increase in Balanced-profile similarity from 44.9 in 2023–24 to 63.4 in 2025–26, while a scoring-weighted specification produces a different temporal pattern. This case is not evidence that the two players have equal value, nor that one will develop into the other. It demonstrates instead that multivariate statistical similarity is conditional on the representation used and can evolve as roles and league-relative profiles change.

The paper makes no claim to a new learning algorithm. Its contribution is an applied, reproducible comparison in which descriptive player representation and supervised forecasting are connected, model selection is separated from final holdout evaluation, a simple persistence benchmark remains visible throughout, and subgroup diagnostics are used to challenge the aggregate result rather than merely confirm it.

# 2 Related Work

Modern basketball analytics includes player evaluation, role characterization, shot modeling, longitudinal production, team strategy, and forecasting. Terner and Franks (2021) review this broader statistical literature and emphasize that player performance can be represented in multiple ways depending on the decision being supported. That perspective is important for PlayerMatch: similarity, clustering, and forecasting are related, but they are not interchangeable tasks.

Player-type analysis has often used multivariate methods rather than traditional position labels. Muniz and Flamand (2022) use clustering and network methods to identify NBA player types across multiple seasons, motivating the idea that functional roles can contain information that five nominal positions do not capture. PlayerMatch adopts the same general motivation but uses a deliberately compact feature set and exposes the representation directly through standardized statistics, weighted cosine similarity, PCA, and k-means.

Forecasting player performance is also an established problem. Vinué and Epifanio (2019) treat basketball careers as sparse longitudinal data and show that future player behavior can be forecast competitively using functional methods. More recent work has compared large collections of machine-learning algorithms for basketball performance forecasting. Papageorgiou et al. (2024) benchmark fourteen methods across advanced basketball statistics and find strong results for tree-based models in their setting, while Papageorgiou et al. (2025) again evaluates multiple model families in individualized NBA fantasy-performance forecasting. These studies show why flexible nonlinear methods are plausible candidates, but they also reinforce that model rankings depend on the target, feature construction, sample, and validation design.

The methodological choices in this paper follow standard statistical ideas. Ridge regression introduces coefficient shrinkage to stabilize prediction with correlated predictors (Hoerl and Kennard, 1970). Gradient boosting constructs nonlinear additive models in a stagewise manner and can capture interactions that a linear model cannot represent directly (Friedman, 2001). PCA provides a low-dimensional linear representation of multivariate variation (Jolliffe, 2002); k-means provides a simple partition of a multivariate space (MacQueen, 1967); and silhouette width is a useful descriptive measure of cluster separation (Rousseeuw, 1987).

Finally, forecasting evaluation must respect temporal order. Random train-test splitting can allow a model-selection process to use information that would not have been available at the time of prediction. Rolling- or expanding-origin designs are standard approaches for evaluating forecasts out of sample (Tashman, 2000). PlayerMatch therefore uses expanding-window validation for model selection and reserves the latest observed season as a final untouched holdout.

# 3 Data

## 3.1 Source and eligibility

The dataset was constructed from the NBA Stats API and contains regular-season player summaries from 2020–21 through 2025–26. The frozen PlayerMatch V5.1 dataset uses a minimum eligibility threshold of 15 games played and 10 minutes per game. This filter is intended to reduce extremely small-sample player seasons while retaining rotation players rather than only starters. The resulting eligible-player counts are 403, 423, 409, 411, 422, and 420 across the six seasons, respectively.

The analysis uses the repository state at commit `dcebfeb81e56276265d2d2bdefc61fd57fa98677`. The code repository is available at `https://github.com/mtnnguyen/PlayerMatch`. Freezing the commit matters because both the player-representation logic and the forecasting artifact are generated code outputs rather than hand-entered tables.

### 3.2 Player representation variables

Each eligible player-season contains ten model features:

| Category | Feature | Symbol used in text |
|---|---|---|
| Scoring | Points per game | PTS |
| Rebounding | Rebounds per game | REB |
| Playmaking | Assists per game | AST |
| Defense | Steals per game | STL |
| Defense | Blocks per game | BLK |
| Shooting | Field-goal percentage | FG% |
| Shooting | Three-point percentage | 3P% |
| Shooting | Free-throw percentage | FT% |
| Role | Usage rate | USG% |
| Efficiency | True shooting percentage | TS% |

PlayerMatch stores per-game, per-36-minute, and per-100-possession versions where applicable. The main analyses in this paper use the per-game basis unless otherwise stated, because the forecasting targets are also season-level per-game outcomes. Rate-adjusted views remain useful in the application for separating role from playing time, but they are not mixed with the forecasting target definition.

### 3.3 Forecasting sample

A forecasting example is created when an eligible player in season $t$ is also eligible in season $t+1$. This produces 1,651 player-season transitions. The supervised feature vector has 21 inputs: the ten current-season model statistics; age, minutes, and games played; six season-over-season changes (PTS, REB, AST, TS%, USG%, and minutes); an indicator for whether the immediately previous eligible season is available; and the number of eligible historical seasons observed for that player.

The five forecast targets are next-season PTS, REB, AST, TS%, and USG%. The transition construction has an important consequence: a player must remain eligible in the following season to become a training example. Players who leave the league, miss most of the next season, or fall below the participation threshold do not contribute an observed target. The supervised sample therefore has survivorship and availability bias. Forecasts should be interpreted as conditional on a player returning and meeting the eligibility definition rather than as unconditional predictions of availability.

## 4 Player Representation and Similarity

### 4.1 Season-relative standardization

Raw box-score quantities are not directly comparable in a multivariate distance calculation because they use different scales. For each season and feature $j$, PlayerMatch standardizes player $i$ as

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \tag{1}$$

where $\mu_j$ and $\sigma_j$ are the eligible-league mean and population standard deviation for that season. This has two useful effects. First, points, rebounds, shooting percentages, and usage can coexist

in the same vector without one feature dominating simply because of its numerical scale. Second, similarity is league-relative. A given raw stat line may occupy a different part of the multivariate space when league distributions change.

### 4.2 Weighted cosine similarity

For an analysis profile with nonnegative feature weights $w_j$, the standardized vector is transformed as

$$\widetilde{z}_{ij} = \sqrt{w_j}\, z_{ij}. \tag{2}$$

The raw pairwise similarity between players $a$ and $b$ is then

$$S(a,b) = \frac{\widetilde{\mathbf{z}}_a^{\top}\widetilde{\mathbf{z}}_b}{\|\widetilde{\mathbf{z}}_a\|\|\widetilde{\mathbf{z}}_b\|}. \tag{3}$$

Using $\sqrt{w_j}$ means the coordinate product in the cosine numerator is multiplied by $w_j$, while the weights also affect both vector norms. A weight of 2 therefore does not mean that a feature contributes exactly twice as much to the final normalized score in every comparison.

The application displays $100S(a,b)$ after clipping to the interval $[0,100]$. Negative cosine values are therefore displayed as zero. This floor is important when interpreting large apparent changes that begin at zero, because the underlying negative value is not visible. The Gradey Dick–De'Andre Hunter case used later in the paper does not rely on that floor: all reported Balanced-profile scores are positive.

### 4.3 Analysis profiles

The Balanced profile assigns weight 1 to all ten features. The Scoring profile increases the influence of points, usage, shooting, and efficiency while reducing rebounding, playmaking, and defensive weights. Playmaking and Defense profiles provide other decision-specific views in the application. These profiles are not claims that one definition of similarity is universally correct. They are explicit assumptions that allow a user to ask different basketball questions while keeping the mathematical mechanism fixed.

A central interpretive rule follows from this design: **similarity is not player value**. Two players can occupy similar league-relative statistical directions while differing greatly in volume, role stability, availability, contract value, defense not captured by the selected box-score variables, or playoff impact. PlayerMatch therefore describes statistical profile alignment rather than ranking one player above another.

### 4.4 PCA visualization

For visualization, PlayerMatch projects the same standardized weighted feature matrix into two dimensions using principal components. PCA is used to summarize dominant directions of variance, not to replace the ten-dimensional similarity calculation (Jolliffe, 2002). A pair that appears close on the first two components is not guaranteed to have the highest full-space cosine similarity, because the two-dimensional projection necessarily discards some information.

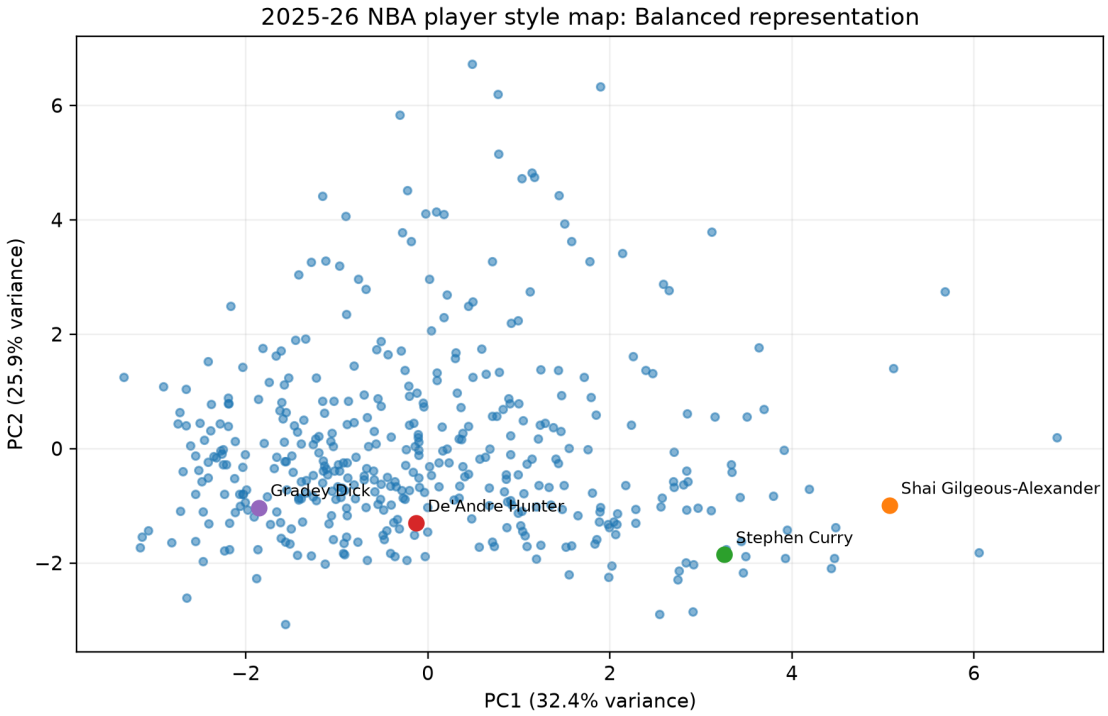



**Figure 1:** PlayerMatch PCA style map for the 2025–26 eligible player pool under the Balanced representation. PC1 explains 32.4% of variance and PC2 explains 25.9%, for 58.3% combined. The two-dimensional map is a visualization of the ten-dimensional standardized feature space; it is not the similarity score itself.

The frozen 2025–26 map contains 420 eligible players. The first two components explain 58.3% of the total variance, meaning that 41.7% remains outside the displayed plane. This is enough for a useful overview of league structure, but not enough to treat visual distance as a substitute for the full representation. The highlighted examples make the point directly: Shai Gilgeous-Alexander and Stephen Curry occupy the high-PC1 side of the map, while De'Andre Hunter and Gradey Dick sit farther left. Dick and Hunter also have similar PC2 coordinates but remain separated along PC1. Their full-space cosine similarity can therefore increase even when the two points are not nearest neighbors on the two-dimensional plot.

### 4.5 Longitudinal similarity example

Because standardization is recomputed within each season, PlayerMatch can also track whether two players move toward or away from a similar league-relative profile over time. Figure 2 previews the paper's main longitudinal example. Gradey Dick and De'Andre Hunter become steadily more similar under the Balanced profile across Dick's three eligible seasons, while the scoring-weighted trajectory is less monotonic. Section 9 returns to this comparison as a case study and separates statistical-profile convergence from claims about player value or future development.

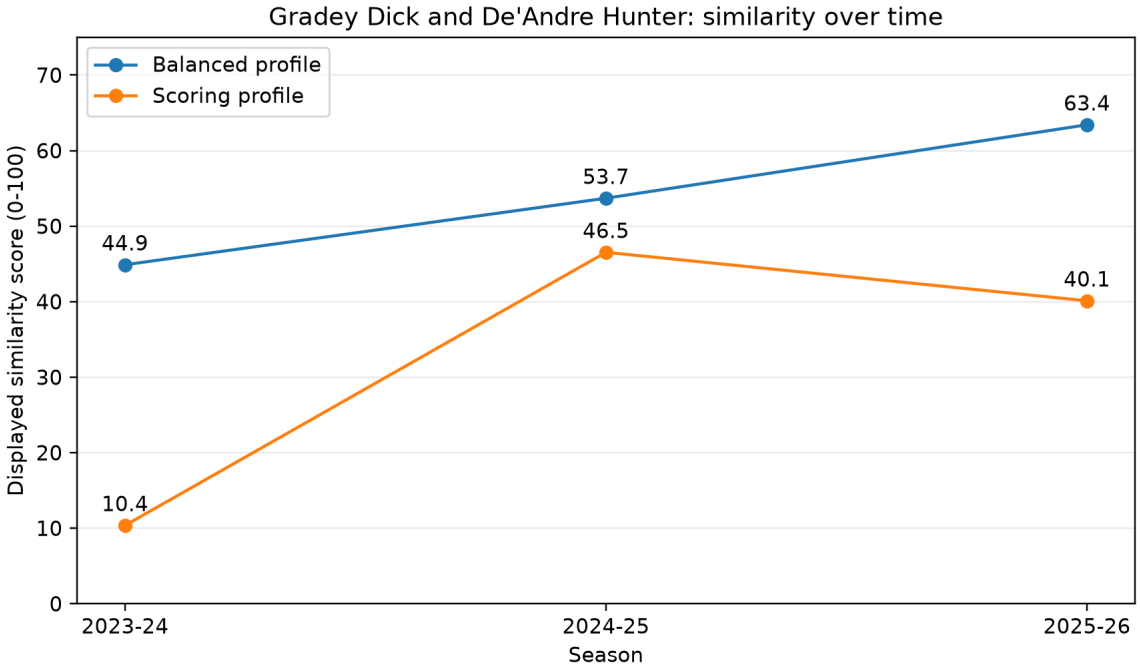



**Figure 2:** Displayed PlayerMatch similarity between Gradey Dick and De'Andre Hunter. Balanced similarity rises steadily, while the Scoring profile increases sharply in 2024–25 before declining in 2025–26.

## 5 Dimensionality Reduction and Archetypes

PlayerMatch also uses k-means clustering as an exploratory way to group players with similar standardized statistical profiles. Clustering is performed in the full ten-dimensional weighted feature space rather than on the two displayed principal components. The PCA map is therefore a visualization of the clustering context, not the space in which cluster assignments are computed.

The implementation evaluates nearby values of $k$ (5, 6, and 7) and reports silhouette separation, with $k = 6$ used as the default interactive view. The silhouette statistic compares within-cluster cohesion with nearest-cluster separation (Rousseeuw, 1987). The initialization is deterministic so that identical data and settings produce identical assignments.

The archetype component is intentionally treated as descriptive rather than as a primary inferential result. A k-means label is not a natural basketball position, and the cluster number itself has no meaning. More importantly for longitudinal analysis, the model is refit independently within each season. A cluster labelled "2" in one season is therefore not automatically the same population as cluster "2" in the next season. A formal longitudinal study would need a cluster-matching procedure, such as centroid assignment across seasons, before making claims about stable archetype lineage.

For this reason, the paper uses clustering to support interpretation of the representation but does not use a cluster label as evidence for the forecasting findings. The more reproducible longitudinal example is the pairwise similarity analysis in Section 9, because the same defined feature weighting can be recomputed within each season and its limitations stated directly.

# 6 Next-Season Forecasting Models

## 6.1 Persistence baseline

The simplest candidate assumes that next season equals the current season:

$$\widehat{y}_{i,t+1} = y_{i,t}. \tag{4}$$

This persistence forecast is intentionally difficult to ignore. NBA season-level statistics often show continuity, and a machine-learning model should not receive credit merely for learning that good scorers tend to remain good scorers. A candidate is useful only to the extent that it improves on what was already available from the current season.

## 6.2 Ridge regression

The linear candidate standardizes the 21 supervised features and fits Ridge regression with $\alpha = 10$. Ridge minimizes squared error with an $L_2$ coefficient penalty and is useful when predictors are correlated (Hoerl and Kennard, 1970). Correlation is expected here: points, usage, minutes, efficiency, and their recent changes are not independent basketball quantities.

Ridge is also attractive for interpretation. It can shrink noisy coefficients while preserving a relatively transparent mapping from the feature vector to the prediction. That does not make its coefficients causal. They describe predictive association after regularization and standardization, conditional on the other modeled variables.

## 6.3 Gradient Boosting

The nonlinear candidate is a Gradient Boosting regressor with 160 estimators, learning rate 0.04, maximum tree depth 2, minimum leaf size 8, and Huber loss. Gradient boosting can represent thresholds and feature interactions that Ridge cannot express directly (Friedman, 2001). It is therefore the natural "more complex" candidate in the experiment.

The comparison is deliberately limited to these three model families. The objective is not to exhaust every possible algorithm. It is to test whether a flexible nonlinear model earns a clear out-of-sample advantage over two strong, interpretable references: persistence and regularized linear regression.

## 6.4 Forecast ranges and error bands

Final player forecasts are clipped to plausible display ranges: 0–60 PTS, 0–30 REB, 0–20 AST, and 0–100 for TS% and USG%. These bounds are safeguards for presentation rather than learned constraints.

PlayerMatch also reports an empirical historical error band equal to the 80th percentile of absolute validation residuals for the selected model, with holdout residuals used only as a fallback if validation residuals are unavailable. This quantity is a descriptive forecast-error summary. It is *not* a formal 80% confidence interval or conformal prediction interval.

# 7 Experimental Design

## 7.1 Temporal validation

For each target, candidate models are compared using expanding-window validation. At a validation target season $T$, training uses only transitions whose target seasons occur before $T$. The model is then evaluated on transitions ending in $T$. The window expands forward and the errors are pooled. This preserves the direction of time and follows the logic of rolling-origin out-of-sample evaluation (Tashman, 2000).

The latest observed target season, 2025–26, is excluded from model selection. After a model family is selected using earlier seasons, it is fit on all pre-holdout transitions and evaluated once on the 2025–26 holdout. Only after that evaluation is the selected family refit on all observed transitions to produce 2026–27 forecasts.

## 7.2 Two-percent simplicity rule

The model-selection rule is fixed before the final holdout is examined. Let $m^*$ be the candidate with the lowest finite expanding-window validation MAE. A candidate is considered practically competitive when

$$\mathrm{MAE}(m) \leq 1.02\,\mathrm{MAE}(m^*). \tag{5}$$

Among candidates inside this 2% tolerance, the least complex model is selected in the order Persistence < Ridge < Gradient Boosting. This is not the one-standard-error rule and is not presented as theoretically optimal. It is a transparent engineering choice: a tiny validation improvement should not automatically justify an additional level of model complexity.

The 2% threshold was not tuned against 2025–26. Changing it after observing the holdout would contaminate the evaluation and make the final error an input to model selection rather than an independent check.

## 7.3 Metrics and subgroup diagnostics

The primary metric is mean absolute error (MAE), with root mean squared error (RMSE) also recorded. Improvement over persistence is reported as

$$100 \times \frac{\mathrm{MAE}_{\mathrm{baseline}} - \mathrm{MAE}_{\mathrm{model}}}{\mathrm{MAE}_{\mathrm{baseline}}}. \tag{6}$$

The holdout is also partitioned using source-season information only. Scoring groups are based on source-season PTS percentile rank within the holdout sample: bottom 10%, middle 80%, and top 10%. Age and minutes groups are also recorded. Mean signed error is defined as

$$\frac{1}{n_G} \sum_{i \in G} (\widehat{y}_i - y_i), \tag{7}$$

so a negative value indicates average underprediction. These subgroup diagnostics are descriptive. No statistical-significance claim is attached to the group comparisons.

# 8 Results

## 8.1 Model selection

All five final targets select Ridge regression. This does not mean Ridge has the smallest validation MAE in every case. For PTS, Gradient Boosting records a validation MAE of 2.2448 compared with 2.2572 for Ridge, a relative difference of only 0.55%. For TS%, the values are 3.2588 and 3.2594, a relative difference of roughly 0.02%. Ridge therefore falls inside the pre-specified 2% simplicity tolerance for both targets and is selected. Ridge is the raw validation-best model for REB, AST, and USG%.

This result directly addresses the initial expectation. The nonlinear model occasionally wins the raw validation comparison, but its advantage is too small and too inconsistent to establish a meaningful complexity benefit under the selection rule.

## 8.2 Final held-out performance

Table 1 reports the untouched 2025–26 holdout. Ridge records lower aggregate MAE than persistence for every target, but the size of the improvement varies sharply.

**Table 1:** Held-out 2025–26 forecasting performance. Lower MAE and RMSE are better. Improvement is measured relative to persistence MAE.

| Target | Ridge MAE | Baseline MAE | Improvement | Ridge RMSE | $n$ |
|---|---|---|---|---|---|
| PTS | 2.634 | 2.843 | 7.35% | 3.365 | 327 |
| REB | 0.814 | 0.865 | 5.95% | 1.052 | 327 |
| AST | 0.733 | 0.746 | 1.67% | 1.022 | 327 |
| TS% | 3.107 | 3.487 | 10.87% | 4.057 | 327 |
| USG% | 2.141 | 2.314 | 7.50% | 2.817 | 327 |

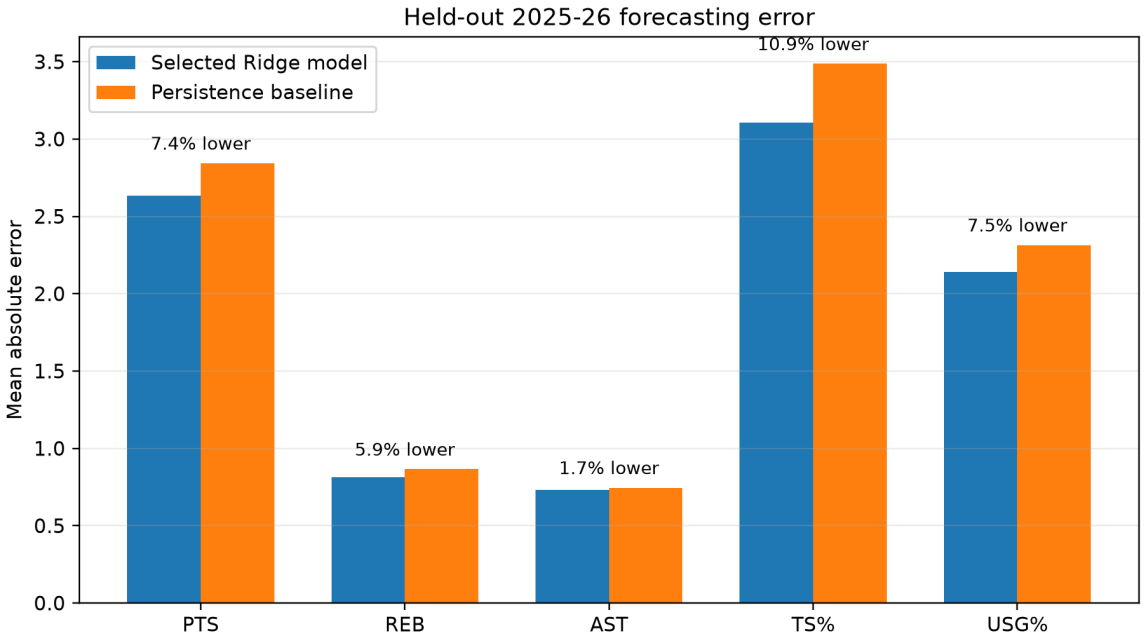



**Figure 3:** Held-out MAE for the selected Ridge model and the persistence baseline. Because the five outcomes use different units and scales, bar heights should be compared within target, while the annotations show relative MAE improvement.

The largest aggregate gain occurs for TS% (10.87%), followed by USG% (7.50%) and PTS (7.35%). REB improves by 5.95%. AST is effectively close to persistence, improving MAE by only 1.67%. It would therefore be misleading to summarize the forecasting system with a single statement such as "the model improves player forecasts." The more accurate conclusion is target-dependent: some season-level outcomes contain useful predictable structure beyond the current-season value, while others provide very little incremental gain under this feature set.

### 8.3 True-shooting subgroup behavior

The aggregate TS% result is the strongest improvement in Table 1, but its subgroup diagnostics complicate that result. For bottom-decile scorers, Ridge MAE is 4.630 percentage points versus 5.485 for persistence, an improvement of 15.58%. For the middle 80%, Ridge records 2.997 versus 3.382, an improvement of 11.37%. Among top-decile scorers, however, the ordering reverses: Ridge MAE is 2.456 while persistence MAE is 2.318. Ridge is 5.96% worse than the baseline in this subgroup.

The direction of the error also matters. The top-scorer group has a mean signed Ridge error of $-1.039$ percentage points, indicating underprediction on average. The bottom group is also negatively biased ($-1.723$), while the middle group is much closer to zero ($-0.269$). These patterns are descriptive rather than inferential, but they demonstrate that a strong aggregate result can hide a meaningful failure mode.

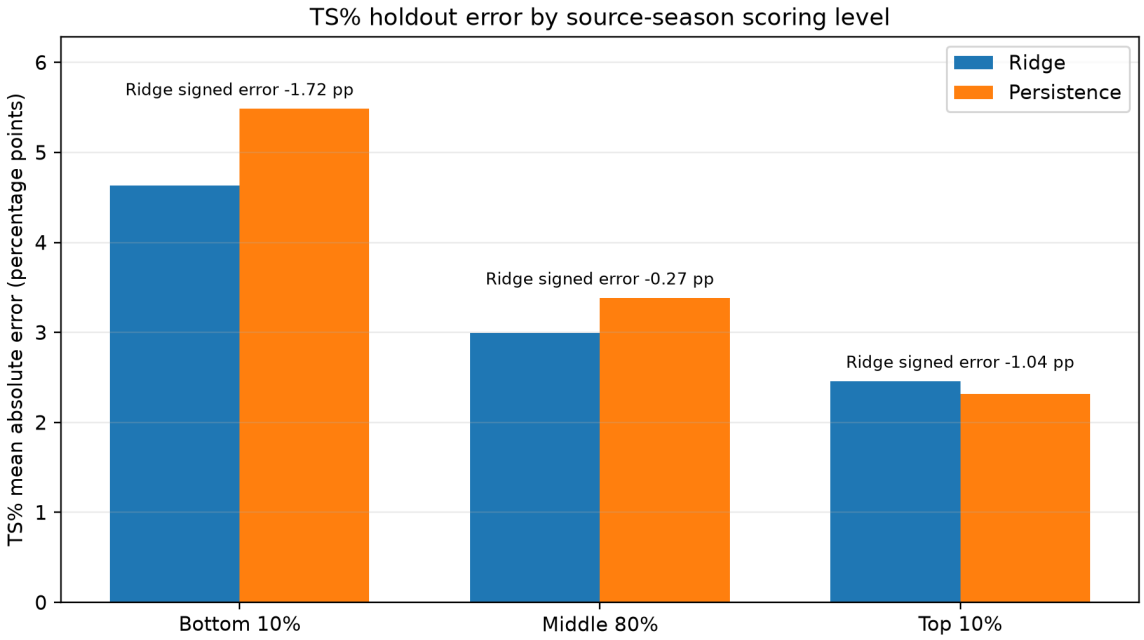



**Figure 4:** True-shooting MAE by source-season scoring group on the 2025–26 holdout. The selected Ridge model improves on persistence for the bottom and middle scoring groups but loses to persistence among top-decile scorers. Annotations report Ridge mean signed error (prediction minus actual).

# 9 Case Studies

## 9.1 Similarity over time: Gradey Dick and De'Andre Hunter

The first case study illustrates how a player-similarity statement depends on the representation. Under the Balanced profile, Gradey Dick's similarity to De'Andre Hunter rises from 44.9 in 2023–24 to 53.7 in 2024–25 and 63.4 in 2025–26. The increase is monotonic across the three eligible Dick seasons.

As shown in Figure 2, the Balanced and Scoring profiles do not tell exactly the same longitudinal story.

The scoring-weighted result tells a different story: 10.4, 46.5, and 40.1. The broad direction from the first to the latest season is still upward, but the peak occurs in 2024–25 rather than 2025–26. That difference is analytically useful. If the comparison were presented only as "Dick became more similar to Hunter," the statement would hide the modeling choice that produced it. The more defensible conclusion is that their overall standardized statistical profiles converged under equal weighting, while the trajectory is less monotonic when scoring volume, usage, and efficiency receive greater influence.

This is not a development projection and not a statement that Dick has become as valuable as Hunter. The similarity measure compares the direction of league-standardized multivariate profiles. A younger player's absolute production can remain lower while his relative pattern across scoring, rebounding, playmaking, defense, shooting, usage, and efficiency becomes more aligned with another player.

The PCA map in Figure 1 provides a useful cross-check on interpretation. Dick and Hunter are not the closest pair in the displayed two-dimensional plane, even though their Balanced cosine similarity reaches 63.4 in 2025–26. This is not a contradiction. The PCA plot retains 58.3% of total

variance, whereas the similarity score is calculated in the full ten-dimensional standardized space. The case therefore also illustrates why PlayerMatch treats PCA as a visualization rather than as the matching model itself.

### 9.2 Forecasting failure: Shai Gilgeous-Alexander

The second case is chosen as an illustration of the top-scorer TS% subgroup result rather than as independent proof of model bias. Shai Gilgeous-Alexander entered the held-out transition with a source-season TS% of 63.7. The Ridge model projected 61.2, while persistence remained at 63.7. His observed 2025–26 TS% was 66.5.

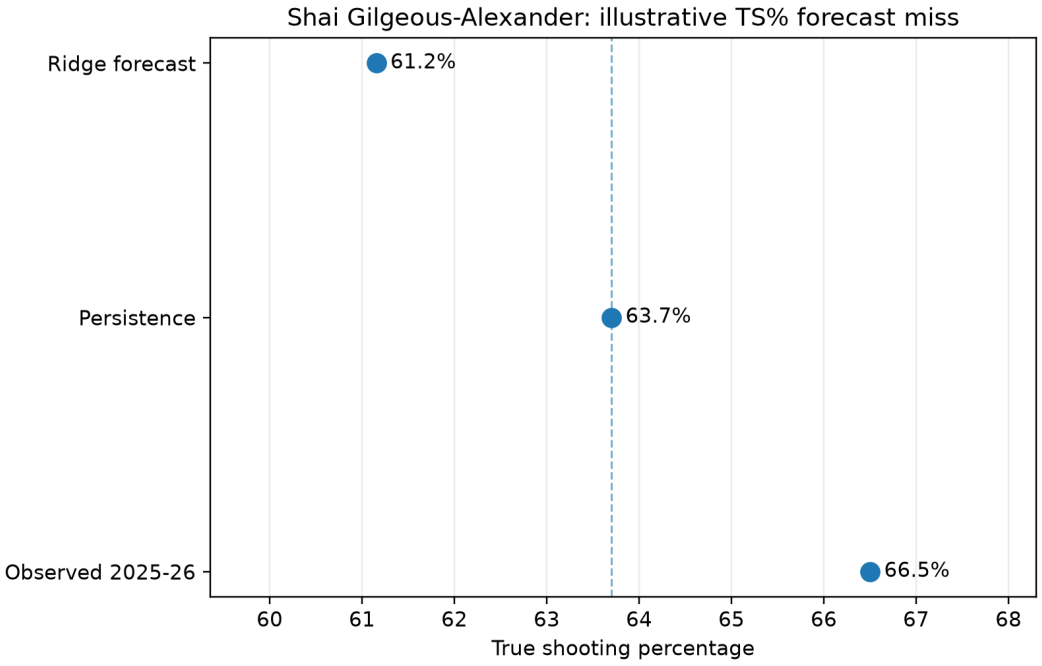



**Figure 5:** Illustrative TS% forecast miss for Shai Gilgeous-Alexander. The Ridge forecast moved downward from a 63.7% source-season value to 61.2%, while the observed next-season value rose to 66.5%.

The resulting Ridge error is −5.35 percentage points, compared with a 2.80-point absolute error for persistence. In other words, the model's attempt to adjust away from the previous season made this particular forecast worse. The example is consistent with a shrinkage-type failure mode: an elite efficiency observation is pulled toward a more typical region of the training data even though the player subsequently remains, and in this case becomes, extremely efficient.

The key evidence is still Figure 4, not Figure 5. Selecting one recognizable player after inspecting errors can create a cherry-picking risk. The case is therefore presented only after establishing that the top-decile subgroup as a whole has worse MAE than persistence and a negative mean signed error.

# 10 Discussion

## 10.1 Complexity did not automatically improve forecasting

The most direct result is that greater flexibility did not produce a reliable enough validation advantage to justify itself. Gradient Boosting has the lowest raw validation MAE for PTS and TS%, but the margins over Ridge are approximately 0.55% and 0.02%. Those are small differences relative to the 2% simplicity threshold. For the other three targets, Ridge is already the validation-best candidate.

This finding should not be generalized into "NBA player development is linear." Ridge being competitive under this experiment says much less than that. The supervised feature set consists primarily of season-level summaries, recent deltas, playing-time information, and age. It does not contain the richer contextual variables through which nonlinear player development may express itself: injury state, shot location, lineup role, on-ball possession patterns, coaching changes, team context, or contract and rotation shocks. A nonlinear model cannot recover interactions in information that is not present.

The result instead supports a narrower principle: *within the representation and historical window used here, additional model complexity did not earn a clear out-of-sample advantage.* That distinction matters because complexity can be attractive for reasons unrelated to demonstrated prediction quality. A portfolio project can easily become a showcase of algorithms rather than a test of whether those algorithms add useful information.

As machine-learning methods continue to become more sophisticated, it is easy to assume that a more complicated model should automatically be better. This experiment did not support that assumption. The conclusion is not that complex models will never outperform simpler ones as methods and data improve. It is that, for this forecasting problem, complexity itself was not evidence of improvement.

## 10.2 Forecastability depends on the statistic

The holdout also shows that "player performance" is too broad to treat as one prediction problem. TS% improves by 10.87% relative to persistence, while AST improves by only 1.67%. The same model family can therefore look useful for one target and nearly unnecessary for another.

One interpretation is that the available predictors capture different amounts of next-season information across outcomes. Recent efficiency, scoring, usage, age, and role variables may help moderate regression and development in TS% and scoring volume. Assists may be more tightly tied to team role, teammates, offensive system, and ball-handling responsibility than the current feature set captures. This is a hypothesis suggested by the error pattern, not a causal conclusion.

The practical implication is that model evaluation should remain target-specific. A single average score across unrelated outcomes would obscure the fact that one target barely improves on persistence while another improves by more than ten percent.

## 10.3 Aggregate performance can hide the decision-relevant failure

The TS% subgroup result is arguably more important than the aggregate improvement. If the analysis stopped at the overall 10.87% gain, TS% would look like the clearest success of the model. Yet the model loses to persistence specifically for top-decile scorers, a group that may be especially relevant in roster, scouting, and player-development decisions.

The Shai Gilgeous-Alexander case makes the mechanism concrete without replacing the subgroup evidence. The model forecasts a decline from an already high efficiency level, while the

observed next season moves higher. Similar individual misses do not necessarily imply a systematic flaw; the subgroup signed error is what makes the case more informative. Even then, the sample has only 33 top-decile scorers, so the finding should motivate further validation rather than a universal correction rule.

A useful future extension would test whether uncertainty or shrinkage should depend on role. For example, a model could condition residual distributions on scoring level or use a hierarchical structure that allows elite high-volume scorers to follow a different transition pattern from low-usage rotation players. That would be a methodologically motivated extension arising from a diagnosed error, rather than feature expansion for its own sake.

### 10.4 Interpretability is not the absence of modeling

The descriptive side of PlayerMatch also reinforces the value of transparent assumptions. The Gradey Dick–De'Andre Hunter case is more useful when both weighting specifications are shown. Balanced similarity increases steadily, while scoring-weighted similarity peaks and then declines. Showing both prevents the similarity score from looking like an objective fact independent of model choice.

This is the broader role of interpretability in the project. The goal is not to avoid models. It is to keep the chain from input statistics to representation to result visible enough that a reader can ask why the result changed.

## 11 Limitations

Several limitations bound the conclusions of this study.

First, the historical window is short. Six observed seasons provide five transition target seasons, and only the seasons before the final holdout participate in model-family selection. This is appropriate for avoiding temporal leakage but limits the diversity of league environments available to the models. A longer history would increase sample size and provide more aging and role-change examples, although older seasons may also be less representative of the current NBA.

Second, the forecasting sample is conditional on next-season eligibility. Because a transition requires the player to appear in both seasons and satisfy the participation thresholds, the model is not trained on the full process that determines whether a player remains available. This creates survivorship and availability bias. The projections are conditional performance forecasts, not injury or roster-survival models.

Third, the feature set is deliberately compact. It does not directly include injuries, transactions, coaching changes, team pace, lineup context, shot quality, spatial tracking, contract status, playoff role, or defensive matchup information. Any of these could explain forecast errors that currently appear as unexplained variation.

Fourth, the 2% simplicity tolerance is a modeling policy rather than a theorem. It is transparent and was not selected using the holdout, but another analyst could reasonably choose a different practical-equivalence threshold. Sensitivity analysis around that threshold would be useful in future work.

Fifth, the historical 80th-percentile absolute-error band is not a formal confidence interval. It summarizes empirical past forecast error and should not be interpreted as having guaranteed coverage for an individual player's future season.

Sixth, the subgroup diagnostics are descriptive. The top-scorer TS% group contains 33 held-out players. The difference between Ridge and persistence in that group is meaningful as a diagnostic, but the analysis does not perform a formal significance test or claim that the same difference must

recur in another season.

Seventh, the similarity model is limited to the ten selected season-level variables. Cosine similarity measures alignment in this statistical representation, not full basketball similarity. Defense in particular is only partially described by steals and blocks. The display also clips negative cosine similarities to zero, which can hide information for very dissimilar pairs.

Finally, the PCA map and k-means archetypes are exploratory. Two-dimensional PCA loses information from the full feature space, and independently refitted k-means labels should not be treated as stable longitudinal identities without an explicit cluster-matching method.

## 12 Conclusion

PlayerMatch began as a player-comparison application but became more useful as a test of a broader modeling assumption: does a more complex algorithm actually improve the decision-relevant result? In this experiment, the answer was not automatically yes.

Across five next-season NBA outcomes, Ridge regression remained competitive with Gradient Boosting during temporal validation and was selected for every target under a pre-defined simplicity rule. On the untouched 2025–26 holdout, Ridge improved on persistence for all five targets in aggregate, but the gain ranged from only 1.67% for assists to 10.87% for true shooting percentage. The strongest aggregate target also contained the clearest subgroup weakness: among top-decile scorers, TS% forecasting was less accurate than persistence and underpredicted on average.

The descriptive analysis leads to a parallel lesson. Gradey Dick and De'Andre Hunter became progressively more similar under the Balanced representation, yet a scoring-weighted representation produced a different trajectory. Similarity, like forecasting, depends on the assumptions encoded in the model.

The main conclusion is therefore methodological rather than algorithmic. More advanced machine learning can be valuable, but complexity should be treated as a hypothesis to test rather than an automatic improvement. For applied sports analytics, a simpler model with clear validation, visible baselines, interpretable assumptions, and honest subgroup diagnostics can be more useful than a more complicated model whose extra flexibility does not translate into meaningful out-of-sample performance.

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# A Reproducibility Notes

## A.1 Frozen implementation

The analyses described in this paper correspond to PlayerMatch V5.1 at commit:

`dcebfeb81e56276265d2d2bdefc61fd57fa98677`

The public repository is `https://github.com/mtnnguyen/PlayerMatch`. The deployed interface is `https://playermatch.vercel.app`.

### A.2 Primary model settings

| Component | Setting |
|---|---|
| Similarity | weighted cosine on within-season z-scores |
| Balanced weights | 1 for all ten features |
| Ridge | StandardScaler + Ridge($\alpha = 10$) |
| Gradient Boosting | 160 estimators, learning rate 0.04 |
| Tree depth | 2 |
| Minimum leaf size | 8 |
| Boosting loss | Huber |
| Validation | expanding window |
| Selection tolerance | 2% of best validation MAE |
| Final holdout | 2025–26 |
| Forecast season | 2026–27 |

### A.3 Figure regeneration

The included helper `scripts/build_paper_figures.py` reads the frozen repository's `public/players.json`, `public/player_history.json`, and `public/projections.json`. When the paper folder is placed beside or inside a local PlayerMatch checkout, the script regenerates the PCA map and the main descriptive and forecasting figures from source data. This avoids inventing PCA coordinates in environments where the full player JSON is not locally available.